RAIN TOMORROW PREDICTOR

A Machine Learning Approach to Next-Day Rainfall Classification on the weatherAUS Dataset

Project Documentation

Machine Learning for Data Analysis Track

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Table of Contents

1. Problem Understanding

This project addresses a binary classification problem: predicting whether it will rain tomorrow (RainTomorrow: Yes/No) at a given Australian weather station, using same-day weather observations such as temperature, humidity, atmospheric pressure, wind, and cloud cover.

1.1 Why It Matters

Short-term rain forecasting derived from routine weather measurements supports everyday decisions in agriculture, transportation, and event planning. A model that reliably flags a likely rain day from standard observations can serve as a lightweight, data-driven complement to full numerical weather prediction.

1.2 Key Framing Decisions

- The target classes are moderately imbalanced (approximately 78% "No" vs 22% "Yes"), so F1-score is used as the primary evaluation metric instead of raw accuracy, and class_weight="balanced" is applied wherever the model supports it.
- RISK_MM (tomorrow's rainfall amount in millimeters) is excluded entirely from the feature set, since it directly determines the label — RainTomorrow is derived from whether RISK_MM exceeds 1mm. Retaining it would constitute data leakage.
- Location is treated as a meaningful categorical feature rather than averaged away, since climate varies substantially across the 49 stations covered.

2. Dataset Understanding

The project uses the weatherAUS dataset, containing daily weather observations from the Australian Bureau of Meteorology.

Property	Value
Rows	142,193
Columns	24
Locations	49 Australian weather stations
Date range	2007-11-01 to 2017-06-25
Numeric features	17
Categorical features	7
Duplicate rows	0

2.1 Column Descriptions

Column	Description
Date	Date of the observation
Location	Australian weather station name
MinTemp / MaxTemp	Minimum / maximum temperature (°C)
Rainfall	Rainfall recorded for the day (mm)
Evaporation	Class A pan evaporation (mm), 24h to 9am
Sunshine	Hours of bright sunshine in the day
WindGustDir / WindGustSpeed	Direction / speed (km/h) of the strongest wind gust in 24h
WindDir9am / WindDir3pm	Wind direction at 9am / 3pm
WindSpeed9am / WindSpeed3pm	Wind speed (km/h) at 9am / 3pm
Humidity9am / Humidity3pm	Relative humidity (%) at 9am / 3pm
Pressure9am / Pressure3pm	Atmospheric pressure (hPa) at 9am / 3pm
Cloud9am / Cloud3pm	Cloud cover (oktas, 0-8) at 9am / 3pm
Temp9am / Temp3pm	Temperature (°C) at 9am / 3pm
RainToday	Was Rainfall > 1mm today? (Yes/No)
RISK_MM	Tomorrow's rainfall amount — excluded from features (data leakage)
RainTomorrow	Target. Will it rain tomorrow? (Yes/No)

3. Data Exploration

3.1 Missing Values

Several columns contain substantial missing data, summarized below:

Column	Missing %
Sunshine	47.7%
Evaporation	42.8%
Cloud3pm	40.2%
Cloud9am	37.7%
Pressure9am	9.8%
Pressure3pm	9.8%
WindGustDir / WindGustSpeed	~7.0%
Remaining columns	< 3%

3.2 Target Distribution

RainTomorrow: 77.6% "No" vs 22.4% "Yes" — a moderate imbalance, not severe enough to justify resampling techniques such as SMOTE. This informed the choice of F1-score as the primary metric and the use of class_weight="balanced" on supported models.

3.3 Categorical Feature Coverage

- RainToday: 77.7% No / 22.3% Yes — mirrors the target almost exactly, since both are derived from the same rainfall threshold on consecutive days.
- Location: 49 stations, each with roughly 3,000-3,400 observations — fairly even coverage, with no single station dominating the dataset.
- WindGustDir: close to uniformly distributed across the 16 compass directions, with a slight lean toward W, SE, and E.

3.4 Correlation Insights

The strongest linear relationships among numeric features are between same-quantity measurements taken at different times of day:

Feature Pair	Correlation
MaxTemp ↔ Temp3pm	0.98
Pressure9am ↔ Pressure3pm	0.96

Feature Pair	Correlation
MinTemp ↔ Temp9am	0.90
Sunshine ↔ Cloud3pm	-0.70

This multicollinearity is expected for weather data and is handled naturally by the tree-based models used later; it is not a concern for the linear models since none of the correlated pairs are perfectly collinear.

4. Data Cleaning

- Dropped RISK_MM entirely — the source of potential data leakage.
- Dropped rows with missing RainTomorrow (the label itself), since a missing target cannot be safely imputed.
- Extracted Year, Month, and Day from the Date column, then dropped the original Date column.
- Applied IQR-based outlier clipping to numeric columns, with the clipping bounds computed from the training set only (to avoid test-set leakage) and then applied to both the training and test sets.

5. Exploratory Data Analysis (EDA)

EDA covered both numeric and categorical features: distribution histograms and boxplots for numeric columns (before and after outlier clipping), count plots for categorical columns (RainToday, Location, WindGustDir), and a correlation heatmap across all numeric features. Key findings are summarized in Section 3 above.

6. Feature Engineering

Four new features were engineered to capture daily variation, computed separately on the training and test sets (post-split) to avoid leakage:

Feature	Formula
TempDifference	MaxTemp − MinTemp
HumidityDifference	Humidity3pm − Humidity9am
PressureDifference	Pressure3pm − Pressure9am
WindSpeedDifference	WindSpeed3pm − WindSpeed9am

7. Data Preprocessing

A single scikit-learn ColumnTransformer handles both feature types end-to-end, so the resulting pipeline is self-contained and can accept raw, unprocessed input at inference time:

7.1 Numeric Features

- Median imputation for missing values
- StandardScaler for feature scaling

7.2 Categorical Features

- Most-frequent imputation for missing values
- OneHotEncoder (with unknown categories ignored, dense output for compatibility with all downstream models)

8. Model Training

Sixteen classifiers were trained and evaluated using an identical train/test split, with F1-score as the primary comparison metric:

8.1 Models Trained

- Linear & Probabilistic: Logistic Regression, Naive Bayes, SVC
- Distance & Tree-Based: KNN, Decision Tree
- Bagging Ensembles: Random Forest, Extra Trees, Bagging, Voting
- Boosting Ensembles: AdaBoost, Gradient Boosting, Hist Gradient Boosting, XGBoost, LightGBM, CatBoost, Stacking

8.2 Handling Class Imbalance

class_weight="balanced" was applied on the models actually trained with it, per the instructor's guidance that this only needed to be done for the models where it was straightforward to apply — not exhaustively across every model.

8.3 Baseline Results (F1 Score, Test Set)

Model	Accuracy	Precision	Recall	F1 Score
CatBoost	0.863	0.764	0.562	0.648
XGBoost	0.859	0.751	0.555	0.639
Stacking	0.859	0.754	0.550	0.636
SVC	0.797	0.532	0.787	0.635
Logistic Regression	0.796	0.530	0.790	0.634
LightGBM	0.859	0.757	0.545	0.634
Hist Gradient Boosting	0.858	0.754	0.544	0.632
Voting	0.858	0.789	0.501	0.613
Gradient Boosting	0.851	0.755	0.499	0.601
Extra Trees	0.855	0.789	0.484	0.600
Random Forest	0.853	0.781	0.476	0.592
KNN	0.842	0.704	0.508	0.590
Bagging	0.842	0.723	0.481	0.578
AdaBoost	0.841	0.718	0.479	0.574
Decision Tree	0.791	0.535	0.527	0.531

Model	Accuracy	Precision	Recall	F1 Score
Naive Bayes	0.657	0.368	0.736	0.490

CatBoost and XGBoost lead the comparison, followed closely by Stacking, SVC, and Logistic Regression. Naive Bayes performs weakest, consistent with its independence assumption between features being unrealistic for weather data, where strong correlations exist (e.g. MaxTemp and Temp3pm).

9. Hyperparameter Tuning

GridSearchCV (5-fold cross-validation, F1 scoring) was applied to the two strongest baseline models.

Model	Best Parameters	Best CV F1 Score
CatBoost	depth=8, iterations=400, learning_rate=0.1	0.6451
XGBoost	max_depth=8, n_estimators=400, learning_rate=0.1	0.6511

XGBoost scored marginally higher on cross-validation (a difference of roughly 0.006, not practically significant), while CatBoost achieved the higher F1-score on the held-out test set. CatBoost was selected as the final model since it performed consistently well across the baseline comparison, and the tuning step confirmed the two models are practically equivalent in performance.

Per the instructor's guidance, exhaustive tuning across every model was not required — tuning was expected only on the best-performing model, with any additional tuning serving purely as a supplementary comparison.

10. Feature Importance & Feature Selection

Feature importance was extracted from the final tuned CatBoost model:

Feature	Importance
Humidity3pm	15.8
Pressure3pm	7.8
WindGustSpeed	7.2
PressureDifference	6.3
Sunshine	5.0
MinTemp	3.1
Rainfall	3.0

Feature	Importance
Humidity9am	3.0
HumidityDifference	2.9
Temp9am	2.9

Humidity at 3pm is by far the most influential predictor, followed by afternoon pressure, wind gust speed, and the engineered pressure-difference feature. No features were removed during Feature Selection, since the full feature set already achieved the highest F1-score — the analysis served to confirm that the engineered features (particularly PressureDifference) meaningfully contribute to the model.

11. Final Evaluation

The final CatBoost model was evaluated on the held-out test set (28,439 records):

Metric	Value
Accuracy	86.3%
F1 Score (Yes)	0.648
ROC AUC	0.899

11.1 Classification Report

Class	Precision	Recall	F1 Score	Support
No	0.88	0.95	0.92	22,064
Yes	0.77	0.56	0.65	6,375

11.2 Confusion Matrix

	Predicted: No	Predicted: Yes
Actual: No	20,961 (TN)	1,103 (FP)
Actual: Yes	2,805 (FN)	3,570 (TP)

The model is highly reliable for "No" days (95% recall) and reasonably effective for "Yes" days (56% recall) given the inherent difficulty of the minority class. An AUC of 0.899 indicates strong overall separability between the two classes.

12. Challenges We Overcame

12.1 Incomplete Preprocessing Pipeline

Issue: Imputation was originally implemented as a manual step outside the scikit-learn pipeline, so the saved model alone could not handle missing values — it depended on data having already been cleaned upstream.

Fix: The pipeline was rebuilt so that imputation, scaling, and encoding all live inside a single self-contained ColumnTransformer, allowing the saved model to accept raw, unprocessed input directly.

12.2 Python 3.14 / Library Compatibility

Issue: The trained model was originally pickled using a scikit-learn version with no available build for Python 3.14 — the version installed on the local deployment machine — meaning the saved model could not be loaded there.

Fix: The final model was retrained using library versions confirmed to support Python 3.14 (scikit-learn 1.9.0, CatBoost 1.2.10), producing an equivalent model (matching performance) that loads and runs correctly in the deployment environment.

13. Model Deployment — Streamlit Application

The final model was saved using pickle and wrapped in a locally-runnable Streamlit web application.

13.1 Application Features

- Interactive sidebar form with inputs grouped by category: General, Temperature, Humidity & Pressure, Wind, and Sky & Rain.
- A live gauge chart displaying the predicted rain probability immediately after submission.
- A five-tier result classification, from "Very unlikely to rain" to "Rain very likely", each with a distinct icon and color.
- Consistent preprocessing: the same IQR clipping bounds derived from training are applied to user inputs at inference time, so predictions stay faithful to the distribution the model was trained on.

13.2 Running the Application Locally

Required files: app.py, rain_prediction_model.pkl, model_meta.json, and requirements.txt, placed together in one folder. The application is launched with:

```
pip install -r requirements.txt
streamlit run app.py
```

14. Conclusion

This project took the weatherAUS dataset from raw, uncleaned weather records through a complete machine learning pipeline: exploration and cleaning of 142,193 observations, engineering of four derived features, a leak-free preprocessing pipeline, training and comparison of sixteen classifiers, targeted hyperparameter tuning, feature importance analysis, and final evaluation of a CatBoost model achieving an F1-score of 0.648 and an AUC of 0.899 on unseen test data. The final model was deployed as a working local Streamlit application, completing the path from data to a usable prediction tool.